CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M.Tech Examination (CE/IT)
December 2012

CE703: Algorithms & Computational Complexity

Date: 20.12.2012,Thursday **Time: 10:00 am to 01:00 pm** **Maximum Marks:70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

(a) How many elements of the input sequence need to be checked on the average, assuming that the element being searched for is equally likely to be any element in the array in linear search? How about in the worst case? Justify your answer. 2

(b) Which of the following recurrence is more costly in terms of running time for large values of n ? Justify your answer with reason. 2

1. $f(n) = 2f(n/2) + n$ 2. $f(n) = f(n-1) + n$

(c) Explain at least two cases when divide and conquer technique is not preferred and why? 3

Q-2

(a) Analysis of algorithm is performed with respect to a computational model called Random Access Machine (RAM). Explain the significance of RAM Model in analysis of algorithm. 3

(b) Solve the following homogeneous recurrence: 5

$$t_n = \begin{cases} n & \text{if } n=0,1,\text{or } 2 \\ 5t_{n-1} - 8t_{n-2} + 4t_{n-3} & \text{otherwise} \end{cases}$$

(c) Prove that 6

1) $f(n) = O(g(n))$ and $g(n) = O(h(n))$ then $f(n) = O(h(n))$.

2) $f(n) = \Theta(f(n))$

3) $f(n) = O(g(n))$ if and only if $g(n) = \Omega f(n)$

OR

(c) How to analyze time and space complexity of recursive algorithm? Explain with suitable example. What are the issues that govern or influence the choice of either iterative or recursive algorithm for solving the problem? 6

Q-3 Attempt Any Two. **14**

(a) Suppose you are choosing between the given two algorithms A and B. Algorithm A solves problems by dividing them into five sub problems of half the size, recursively solving each sub problem, and then combining the solutions in linear time. Algorithm B solves problems of size n by dividing them into nine sub problems of size n=3, recursively solving each sub problem, and then combining the solutions in $O(n^2)$ time. What are the running times of each of these algorithms (in big-O notation), and which would you choose? Why? 7

(b) Solve the Single Source Shortest path problem for the diagraph with the weighted matrix given below using greedy approach. Use node A as Source node. Prove that your algorithm is greedy. 7

	a	b	c	d
a	0	∞	3	∞
b	2	0	∞	∞
c	∞	7	0	1
d	6	∞	∞	0

(c) 1. Solve the following recurrence equations: 3

$T(n) = 4T(n/2) + n^2\sqrt{n}$.

2. Solve following recurrence using recursion-tree method: 4

$T(n) = 2T(n/2) + n/\lg n$

SECTION-II

Q-4

(a) Compare dynamic programming with memorization. How memorization can provide efficiency to usual dynamic programming approach? 2

(b) Explain the terms: 2

i) Articulation Point

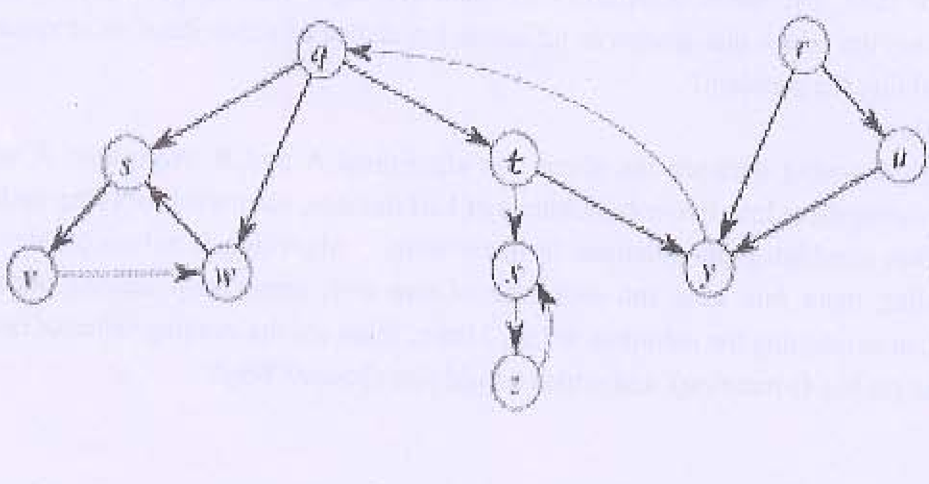
ii) Optimal Substructure property

(c) Working modulo q = 11, how many spurious hits does the Rabin-Karp matcher encounter in the text T = 3141592653589793 when looking for the pattern P = 26? 3

Q-5

(a) List and explain the significance of four factors that affects the efficiency of the backtracking algorithms. 4

(b) Show how Depth-First Search (DFS) works on the graph of following figure. Assume that the DFS procedure considers the vertices in alphabetical order, and assume that each adjacency list is ordered alphabetically. Show the discovery and finishing times for each vertex, and show the classification of each edge. 4



(c)	Explain how to determine the occurrences of pattern P in the text T by examining the π function in KMP Matcher algorithm. Derive the complexity of the algorithm and compare it with Rabin-Karp algorithm for string matching.	6
	OR	
(c)	Distinguish the problems between P and NP and justify why the problem can be identified as P and NP problem? i) Shortest Vs Longest path ii) Euler tour Vs Hamiltonian cycle iii) 2-CNF satisfiability Vs 3-CNF satisfiability	6
Q-6	**Attempt Any Two.**	**14**
(a)	Using Dynamic programming, find an optimum order of a matrix chain product whose sequence of dimensions is <5,10,3,5,6,2>.	
(b)	Let s1 and s2 be two strings of characters. If we want to transform s1 into s2 with the smallest number of operations of the following types: 1) delete a character 2) insert new character 3) change/replace a character Write a dynamic programming algorithm that finds the minimum number of operations needed to transform s1 into s2. What is the complexity of your algorithm?	
(c)	Consider n=4,(w1,w2,w3,w4) = (2,5,10,5) and (p1,p2,p3,p4)=(40,30,50,10) and capacity W= 16. Find the optimal solution using backtracking for 0/1 knapsack problem using given data.	